United International University (UIU)
Department of Computer Science and Engineering
CSE 1325: Digital Logic Design, Midterm Spring 2026
Total Marks: 30, Duration: 1 hour 30 minutes

[Any examinee found adopting unfair means including copy from another examinee will be expelled from the trimester/program as per UIU disciplinary rules.]

Answer All Questions

1. a) In a digital kingdom, a robot accountant named Byte keeps track of treasure using binary arithmetic. One day, the king says: **[3]**

"Byte, we collected 617_{10} gold coins from taxes, and later received another 569_{10} coins from a neighboring kingdom. How many coins do we have in total?"

Byte decides to solve this using BCD addition. Help Byte complete his task by converting the numbers to BCD and performing the BCD addition.

b) In a futuristic digital vault, treasures are secured using numbers written in an unknown base r. One day, the vault displays a secret code involving another digit y, which depends on the base. The system provides the following clues: **[3]**

(i) y = r - 2

(ii) $24_r + 13_r = (3y)_r$ [Note: 3y is a whole number, where 3 and y are separate digits]

To unlock the vault, you must determine the correct values of r and y. Find the values of r and y by interpreting the numbers in base.

2. a) Simplify the following Boolean Expression (using Boolean algebra) to an expression containing minimum number of literals. You must state the identity/law used at every step. **[3]**

$$\overline{(\overline{AB} + BC\overline{D}) + \overline{B}(\overline{A} + C)D}$$

b) Convert the following expression into both canonical SoP (SOM) and canonical PoS (POM) forms using Boolean algebra. **[3]**

$$F(A, B, C, D) = B(A + \overline{D}) + \overline{A}C + BC\overline{D}$$

3. Construct a logic function that detects non-prime numbers in a 4-bit binary number (A,B,C,D), where A is the most significant bit and D the least. A number is considered non-prime if it is 2, 4, 6, 8, 9, 10, 12, 14, 15 (ignore 0 and 1).The output should be 1 for non-prime inputs and 0 otherwise.

(i) Construct the truth table and map the minterms onto a K-map. **[2]**
(ii) Find all prime implicants and essential prime implicants. **[2]**
(iii) Derive the optimized SOP equation for the output using the selection rule. **[2]**

4. Consider the following Boolean function

$$F\ (A,\ B,\ C,\ D) = \prod\nolimits_{M}(3,\ 4,\ 9,\ 11,\ 12) + \sum\nolimits_{d}(2,\ 10,\ 13,\ 14)$$

Provide the optimized function

(i) Using K-map in **[1]**
(ii) Simplified sum-of-products (SOP) and **[2]**
(iii) Simplified product-of-sums (POS) form. **[2]**
(iv) Between minimized SOP and POS, which form will you choose to implement the function and why? **[1]**

5. A high-security system is being developed for the Galactic Republic's vault. The system uses four input signals to determine whether to open the vault door.

The inputs consist of three authentication methods and an override switch:

- Fingerprint scan
- Retina scan
- Access card
- Security override switch

If the inputs are valid, then it is 1; otherwise, it is 0. The switch can be turned ON (1) or OFF (0).

The output 1 indicates that the vault door opens, and 0 indicates that it remains locked.

The vault door opens under the following conditions:

- When all three authentication methods are valid
- When any two of the three authentication methods are valid, and the override switch is OFF
- When the override switch is ON, and at least one authentication method is valid

In all other cases, the vault remains locked.

(i) Construct the complete truth table for the system. **[2]**
(ii) Using a 4-variable Karnaugh map, simplify the Boolean function for the output. **[3]**
(iii) Draw the logic circuit using basic gates (AND, OR, NOT). **[1]**